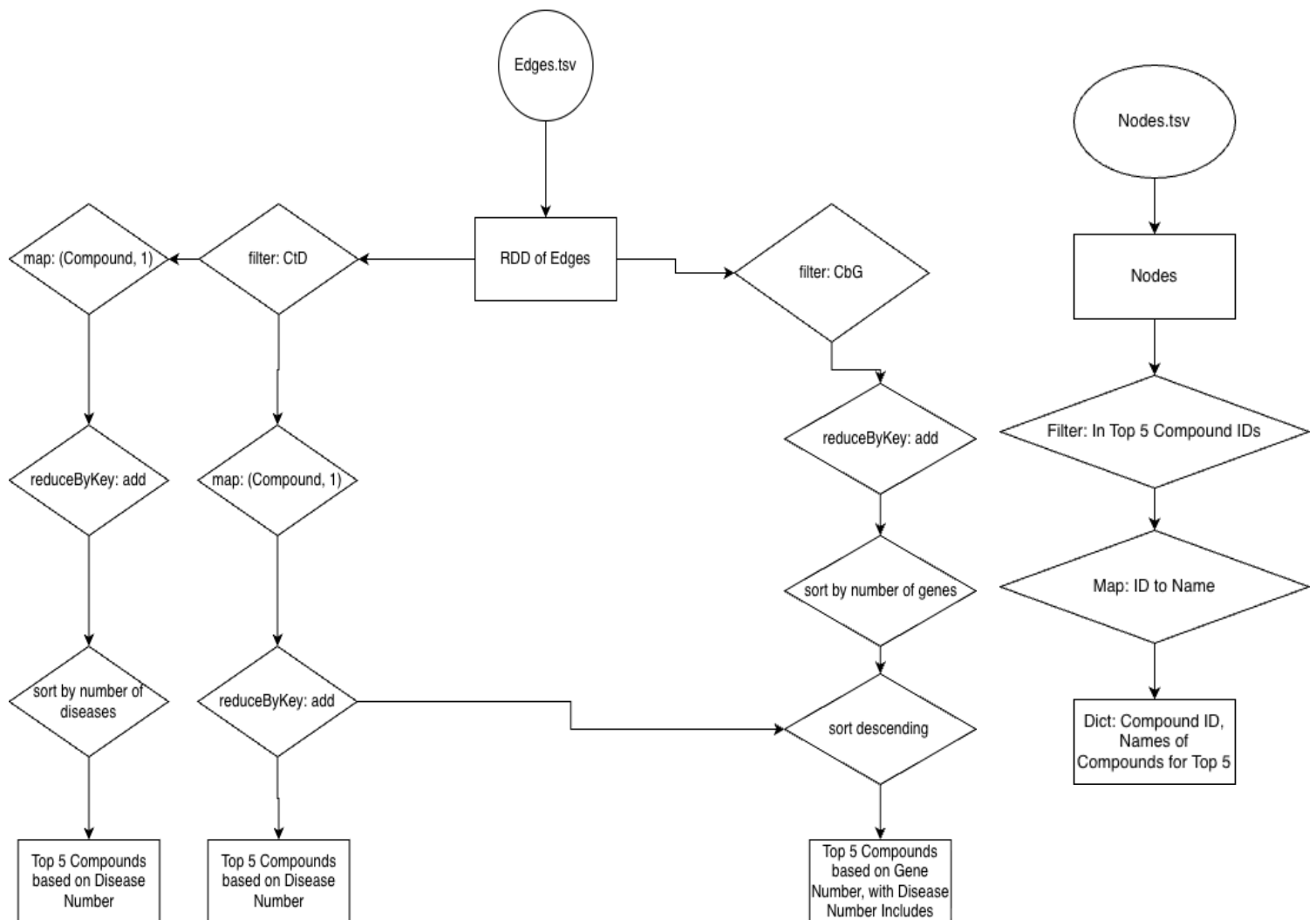


I've provided two flowcharts to describe my process. The first covers the use of filters and maps for Edges.tsv to find the Top 5 for all queries. The second chart reveals how I created the dictionary to convert the IDs into names for my output. I did not include my use of collect() or collectAsMap(), but it was used for every return.



Here are my results for Q1, Q2, and Q3 in order:

Sunitinib, Number of Genes: 132, Number of Diseases: 2

Bosutinib, Number of Genes: 104, Number of Diseases: 0

Crizotinib, Number of Genes: 85, Number of Diseases: 1

Dasatinib, Number of Genes: 64, Number of Diseases: 1

Metformin, Number of Genes: 56, Number of Diseases: 3

Methotrexate -> 19 diseases

Doxorubicin -> 17 diseases

Prednisone -> 15 diseases

Epirubicin -> 14 diseases

Prednisolone -> 13 diseases

Sunitinib -> 132 genes

Bosutinib -> 104 genes

Crizotinib -> 85 genes

Dasatinib -> 64 genes

Metformin -> 56 genes